KALEB MARTINY

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EDUCATION

University of California, San Diego | M.S. Aerospace Engineering Expected June 2027
- Incoming graduate student focused on propulsion, aerodynamics, and systems design

Illinois Institute of Technology | B.S. Aerospace Engineering | GPA: 3.74 May 2026
- Honors/Activities: Magna Cum Laude, Tau Beta Pi, Dean's List, IIT Rocketry Club/SEDS Chapter

Wheaton College | B.S. Liberal Arts Engineering | GPA: 3.40 May 2025
- Honors: Dean's List; Completed Wheaton–Illinois Tech 3-2 Dual Degree Engineering Program

ENGINEERING PROJECTS & EXPERIENCE

HEIMDALL L4 Satellite Mission | Senior Design Project, Propulsion Lead
Illinois Institute of Technology | 10-Person Team Jan. 2026 – May 2026
- Led propulsion subsystem development for a 10-person team proposing HEIMDALL, a Sun-Earth L4 heliophysics and communications mission for the AIAA Undergraduate Space RFP
- Sized station keeping propulsion for a 15-year Sun-Earth L4 mission, using a 2.0 m/s annual station keeping budget and converging on ~54.35 m/s total mission delta-V with the inclusion of solar radiation pressure
- Selected a Northrop Grumman MRE-5.0 monopropellant system with 232 s Isp, two 22 N thrusters, 19.01 kg propellant mass, 0.121 kg helium pressurant, and 13 min 22 s total impulse time
- Supported spacecraft level trade studies for a Falcon 9 launched architecture with a 300 km parking orbit, 5-day checkout, 2-year L4 transfer, X-band/laser communications suite, SWaP allocation, and mission CONOPS

Crop Duster Aircraft Concept | Senior Design Project, Team Lead
Illinois Institute of Technology | 5-Person Team Aug. 2025 – Dec. 2025
- Led a 5-person senior design team developing a 17,428 lb gross-weight agricultural aircraft with 19,000 lb MTOW and 600 nmi range-with-loiter capability
- Performed sizing trade studies for a 77.1 ft span, 758.93 ft² wing, AR = 7.83, takeoff W/S = 23.97 lb/ft², and takeoff T/W = 0.32 configuration
- Conducted XFLR5 analysis of a NACA 4415 wing at 92 m/s cruise speed to generate aerodynamic coefficients and control derivative inputs for flight modeling
- Developed a MATLAB/Simulink 6-DOF flight model to evaluate glide, climb, holding, and landing performance, including L/D = 22.46 in simulated glide and 905.5 ft/min climb rate from 1,500 to 3,500 ft AGL

Engineering Project Management Intern
Wheaton College May 2025 – Aug. 2025
- Coordinated mechanical, electrical, and facilities engineering tasks for campus HVAC and infrastructure projects, supporting design review, verification, compliance review, and technical documentation
- Managed daily progress documentation for a $10M facilities project and developed cost estimates for campus-wide heating and cooling energy optimization initiatives

Particle Turbulence Apparatus Design | Research Assistant
Illinois Institute of Technology Aug. 2024 – Mar. 2025
- Designed structural and flow components of a custom turbulence test apparatus using SolidWorks
- Developed a particle-flow delivery system based on particle aerodynamics and fluid dynamics principles that allows for variable flowrates

ADDITIONAL TECHNICAL EXPERIENCE

Teaching Assistant | Statics, Dynamics, Mechanics of Materials
Wheaton College Aug. 2024 – May 2025
- Supported course instruction in Statics, Dynamics, and Mechanics of Materials by grading students' computational homework and providing feedback to strengthen students' engineering problem-solving methods

Engineering Design & Lab Assistant
Wheaton College Aug. 2023 – May 2024
- Instructed students on safe machine shop operation and supported engineering design projects through prototyping, manufacturability, and equipment guidance

SKILLS

CAD & Modeling: SolidWorks, Fusion 360, AutoCAD, Shapr3D | **Programming & Analysis:** MATLAB, Simulink, XFLR5, Excel
Fabrication & Testing: Additive Manufacturing, Material Testing, Machine Shop | **Methods:** Systems Integration, Design Verification, Experimental Setup, Documentation